YAKEEN

NEET 2.0 2027

Physics | Lecture - 02

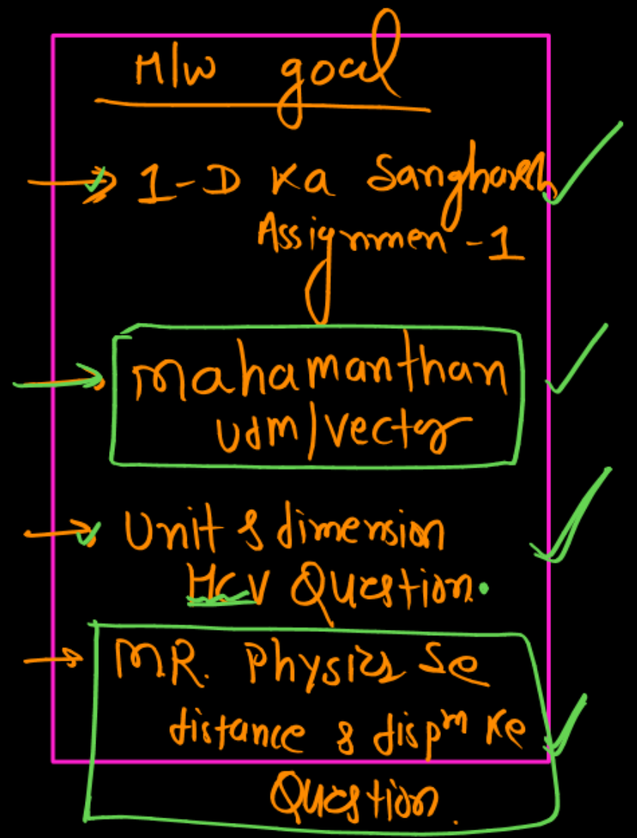
Motion in a Straight Line

By- Manish Raj (MR Sir)



Todays Goal

PhD on distance & displacement.



Question

(12)

TRUE / FALSE

1. Distance must be equal to magnitude displacement in ID motion.
2. If displacement of particle is zero, then distances must be zero.
3. If distance of particle is zero, then displacement must be zero.
4. Magnitude of displacement is always equal to distance.
5. Magnitude of displacement is always equal to distance in ID motion.

Question

Fill in the blanks : (may/must/must not/may not)

A. Distance travelled by object is zero the displacement _____ be non-zero.

B. Distance travelled by object is not zero then displacement _____ be zero.

C. Displacement moved is zero then distance _____ or _____ zero

D. If displacement is not equal to zero then distance _____ equal to zero.

Question

True / False

A. Particle is moving on straight line, distance is equal to displacement ☒

B. Displacement independ of choice of frame of reference ☒

C. For a particle moving on parabolic path distance always grater than displacment ☑

D. Displacement is not zero; then distance must not be zero ☐

E. Distance is not equal to zero; then displacement may equal to zero. ☐

Question

Statement-1: Distance and displacement is different physical quantity.

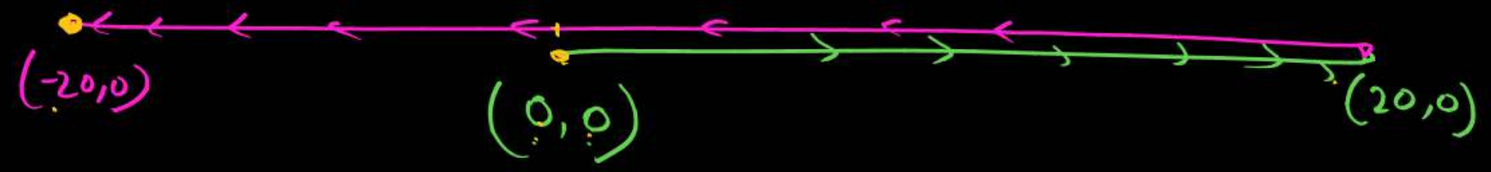
Statement-2: Distance and displacement have same unit and dimension.

1. Both are wrong
2. Both are correct
3. 1st wrong, 2nd correct
4. 1st correct, 2nd wrong

Question

MCV

A particle starts from the origin, goes along the X-axis to the point (20m, 0) and then returns along the same line to the point (–20 m, 0). Find the distance and displacement of the particle during the trip.



$dist^n = 60\,m$

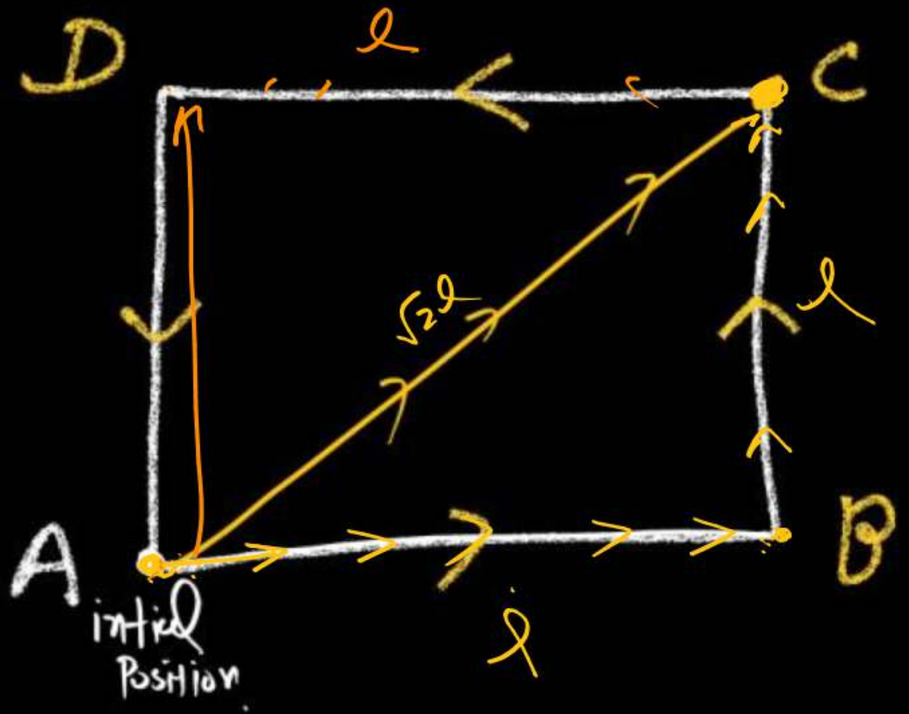
$$disp^m = \vec{r}_f - \vec{r}_i$$

$$= -20\hat{i} - 0$$

$$= -20\hat{i}$$

(13) find distⁿ & dispⁿ; object is moving on square park (ℓ)

Copy

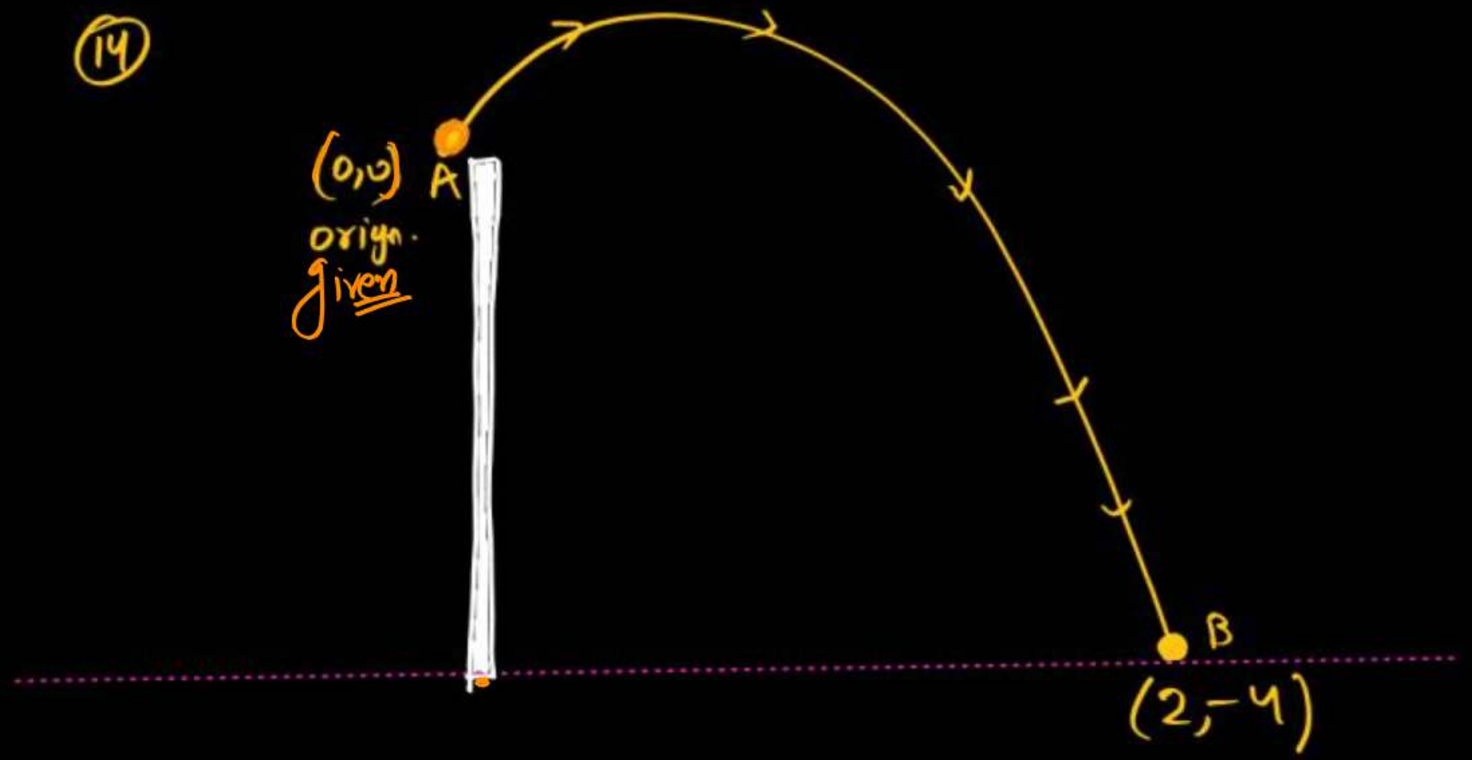


Path	distance	dispⁿ
A → B	ℓ	ℓ
A → C	2ℓ	√2ℓ
A → D	3ℓ	ℓ
A → A	4ℓ	0

$$\text{disp}^n_{Ac} = \ell\hat{i} + \ell\hat{j}$$

$$|\text{disp}^n|_{Ac} = \sqrt{2}\ell$$

(14)



dist^n → (an' find

$$\text{disp}^m = \vec{r}_f - \vec{r}_i$$

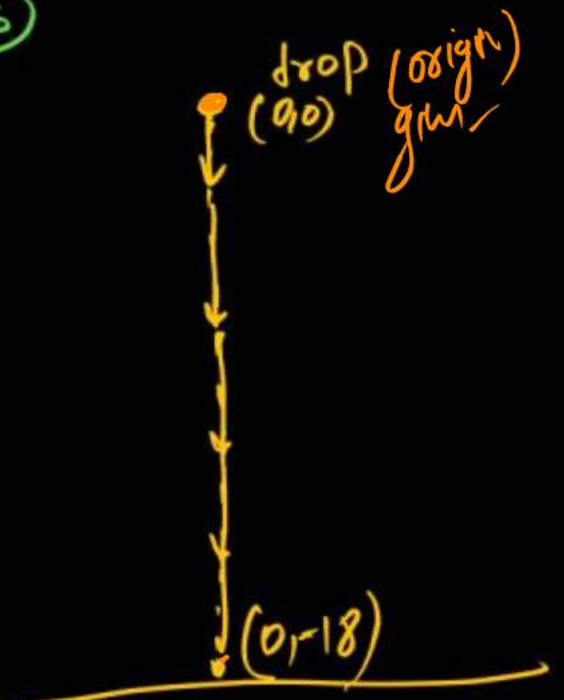
$$= 2\hat{i} - 4\hat{j} - 0\hat{i} - 0\hat{j}$$

$$\vec{s} = 2\hat{i} - 4\hat{j}$$

$$|\vec{s}| = \sqrt{(2)^2 + (-4)^2}$$

$$= \sqrt{20} = 2\sqrt{5}$$

(15)

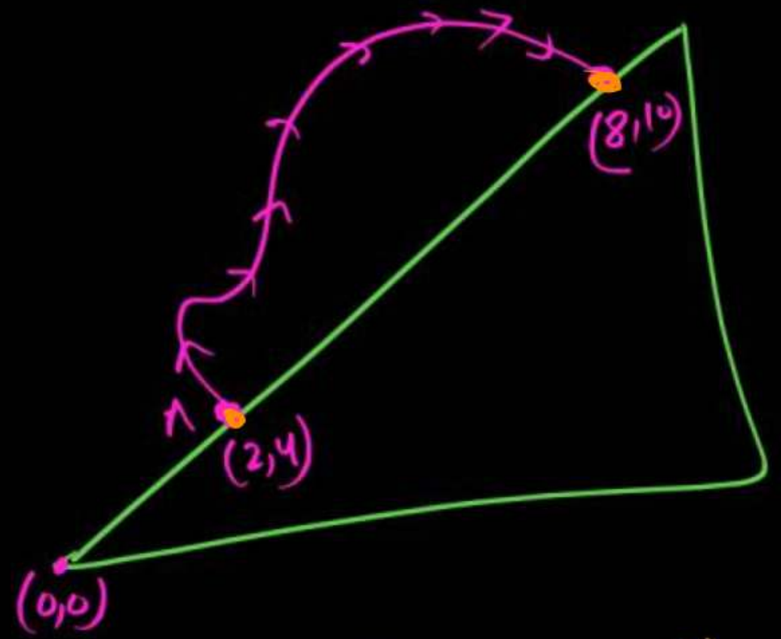


$$\text{disp}^n = -18\hat{j} - 0\hat{i} - 0\hat{j}$$

$$\boxed{\text{disp}^n = -18\hat{j}}$$

$$\text{dist}^n = 18\ m$$

(16)



$$\text{disp}^n = (8i+10j) - (2i+4j)$$

$$= 6i+6j$$

$$|\text{disp}| = 6\sqrt{2}$$

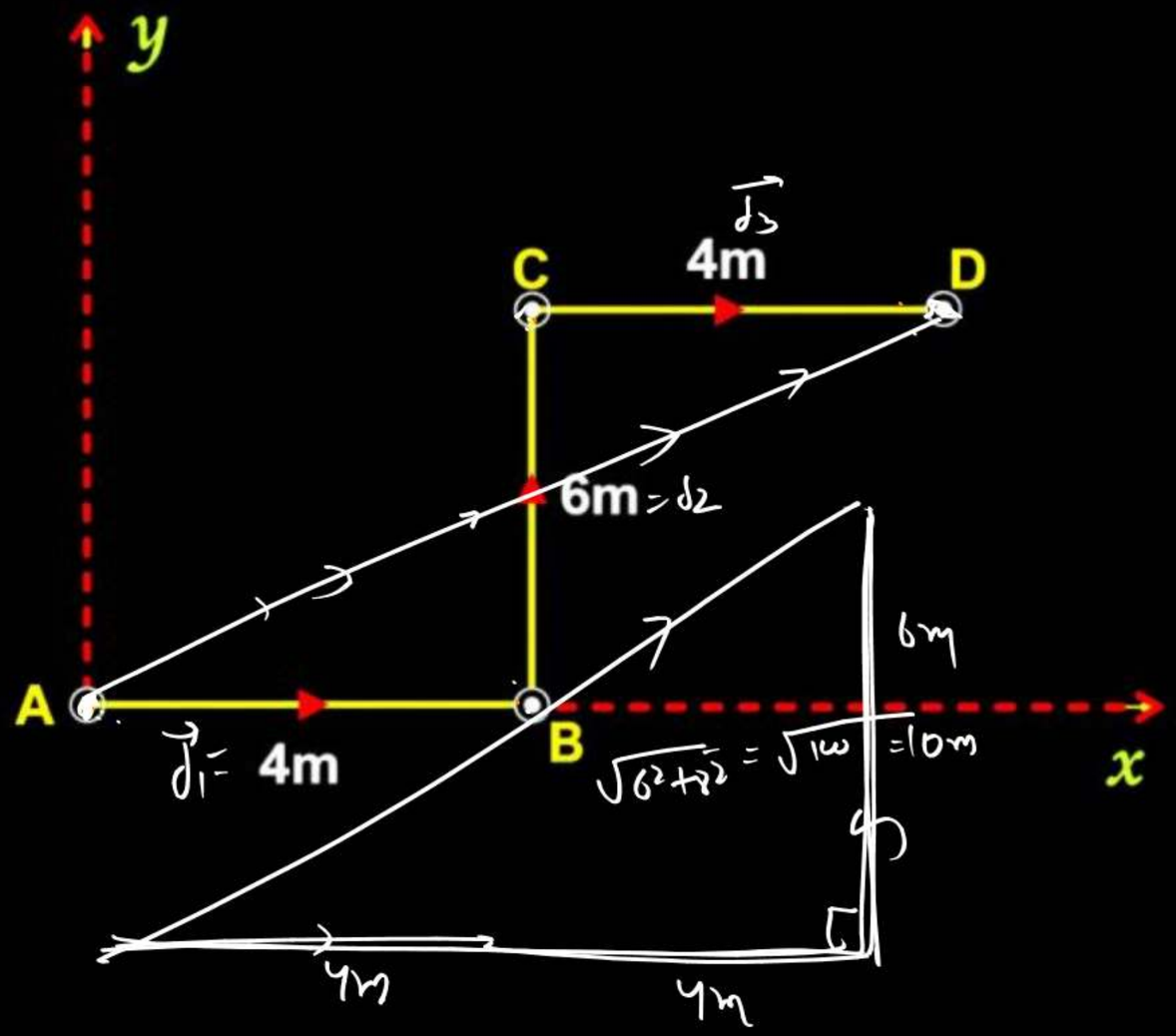
Question 17

Distance and displacement will be?

1. 14m, 8m
2. 14m, 6m
3. 14m, 10m
4. None

$\text{disp}^{m} = \vec{r}_f - \vec{r}_i$

$\vec{\text{disp}} = \vec{d}_1 + \vec{d}_2 + \vec{d}_3$



Question (18) No copy

Correct statement among the following is:

1. When displacement is zero, distance travelled is not zero.
2. When displacement is zero, distance travelled is also zero.
3. When distance is zero, displacement is not zero.
4. Distance travelled and displacement are always equal.

Question 19 (OPY)

If a particle moves from point P(2, 3, 5) to point Q(3, 4, 5). The displacement vector be:

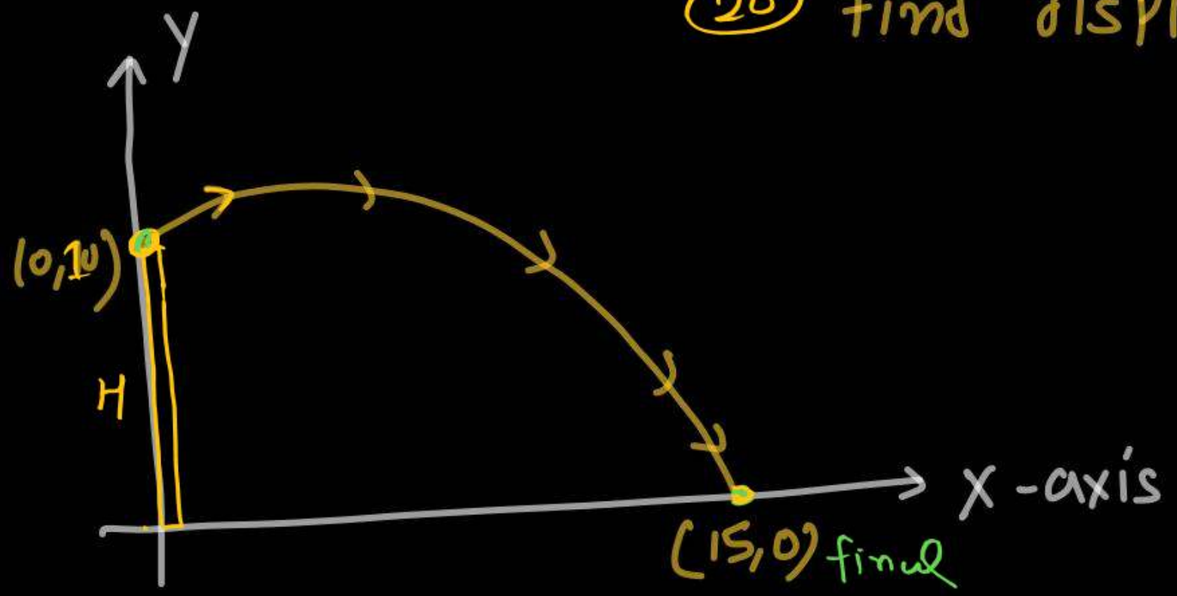
initial — P(2, 3, 5)

final — Q(3, 4, 5)

1. $\hat{i}+\hat{j}+10\hat{k}$
2. $\hat{i}+\hat{j}+5\hat{k}$
3. $\hat{i}+\hat{j}$ ✓
4. $2\hat{i}+4\hat{j}+6\hat{k}$

$$\text{disp}^{n} = \vec{r}_f - \vec{r}_i$$
$$= (3i+4j+5k) - (2i+3j+5k)$$
$$= i+j+0$$

(20) find displacement

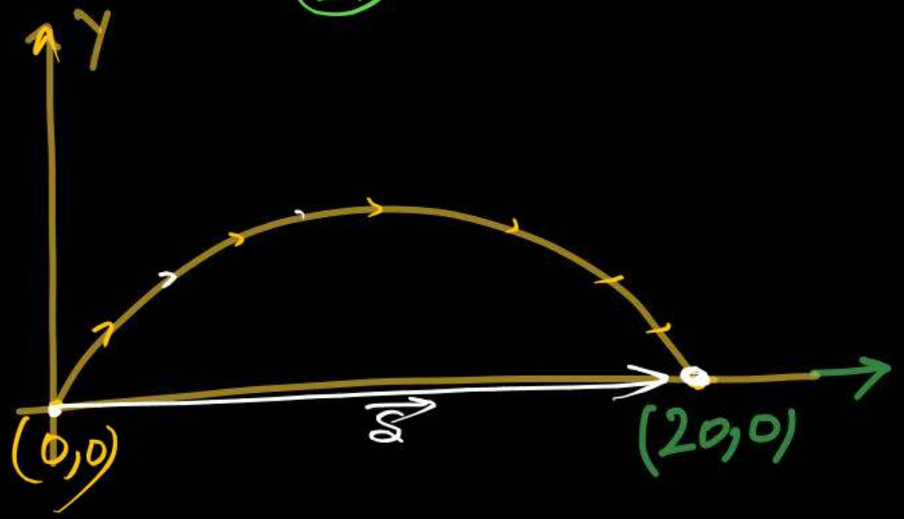


$$\text{disp}^m = (15\hat{i} + 0\hat{j}) - (0\hat{i} + 10\hat{j})$$

$$= 15\hat{i} - 10\hat{j}$$

$$|\text{disp}^m| = \sqrt{(15)^2 + (10)^2}$$

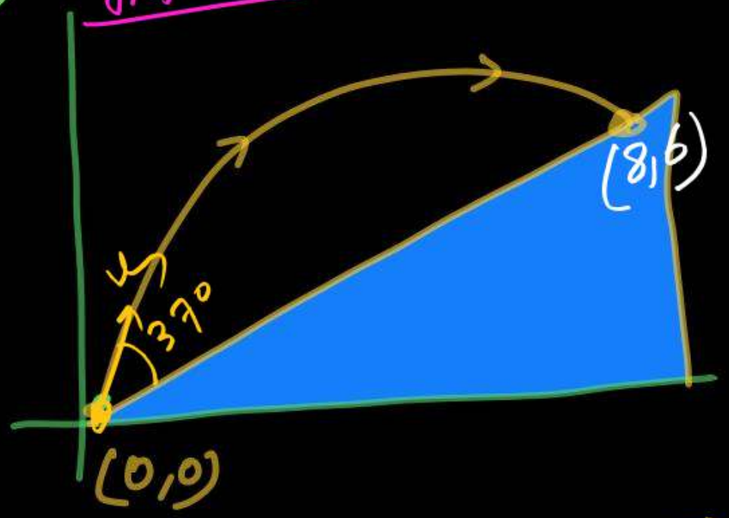
(21)



$$\text{disp}^n = (20\hat{i}+0\hat{j}) - (0\hat{i}+0\hat{j})$$

$$= \underline{20\hat{i}}$$

(22) $\text{disp}^n = ??$



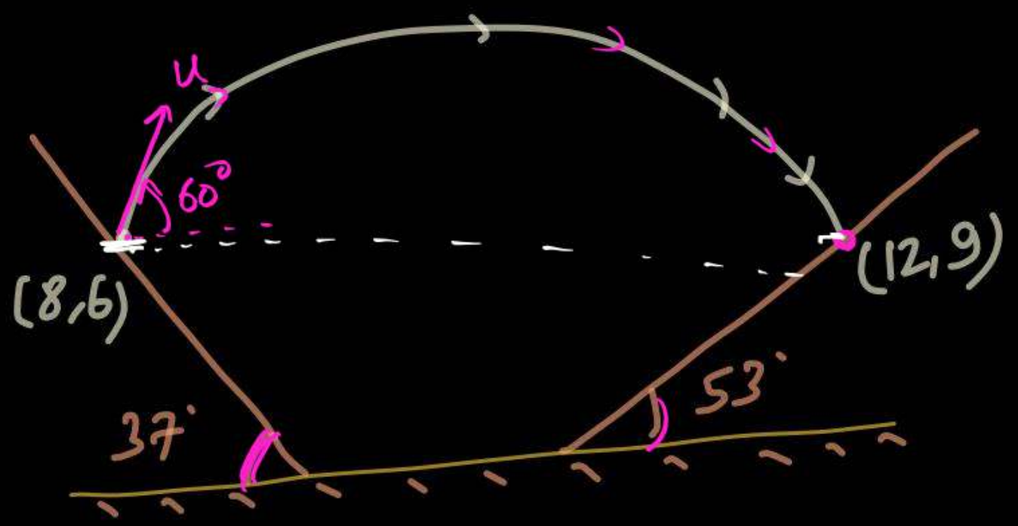
$$\text{disp}^n = \vec{r}_f - \vec{r}_i$$

$$= 8\hat{i}+6\hat{j} - 0\hat{i} - 0\hat{j}$$

$$= 8\hat{i}+6\hat{j}$$

$$|\text{disp}^n| = 10\text{m}$$ ✓

(23)



$$\text{disp}^m = (12\hat{i}+9\hat{j}) - (8\hat{i}+6\hat{j})$$

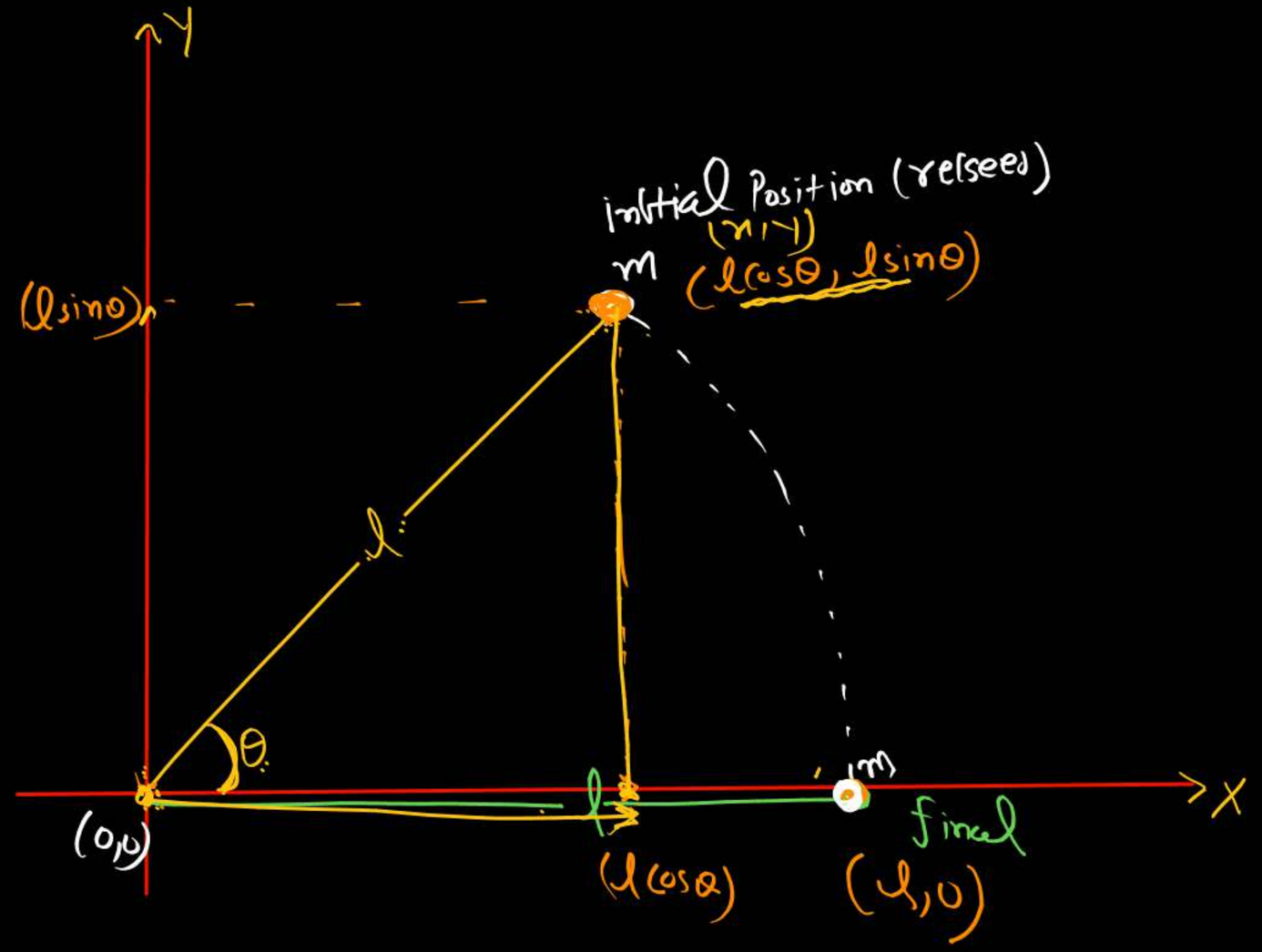
$$= 4\hat{i}+3\hat{j}$$

$$|\text{disp}^m| = \sqrt{4^2+(3)^2}$$

$$= \sqrt{16+9}$$

$$= 5\,m$$

(24) find dispm of object??

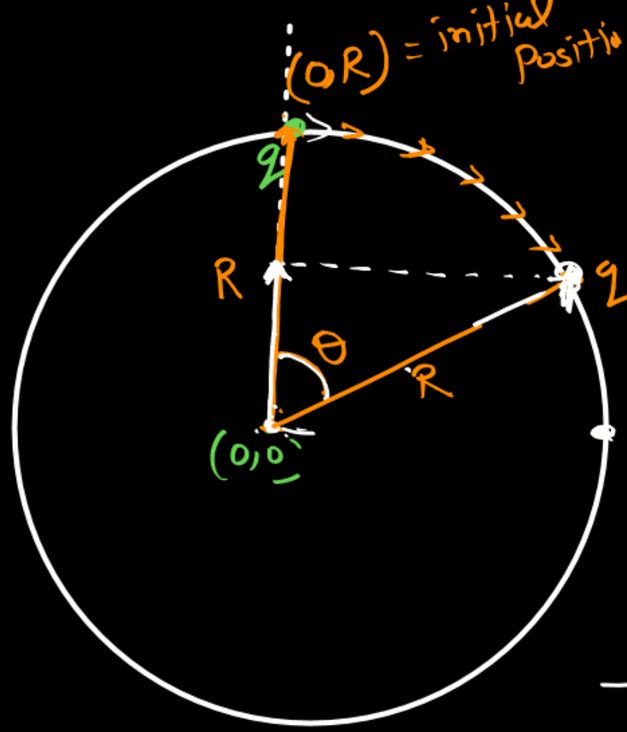


$$\vec{s} = \vec{r_f} - \vec{r_i}$$

$$\vec{s} = (l\hat{i} + 0\hat{j}) - (l\cos\theta\,\hat{i} + l\sin\theta\,\hat{j})$$

$$\vec{s} = l(1-\cos\theta)\,\hat{i} - l\sin\theta\,\hat{j}$$

(25) Charge is moving on circular Path then find its displ^m after Rotating by θ angle: —



$q(x,y) = (R\sin\theta, R\cos\theta)$

$\vec{s} = \vec{r}_f - \vec{r}_i$

$= (R\sin\theta\, i + R\cos\theta\, j) - R\,j$

$\vec{s} = R\sin\theta\, i + R(\cos\theta - 1)\, j$

If $\theta = 60^\circ$

$\vec{s} = R\sin 60\, i + R(\cos 60 - \frac{1}{2})\, j$

$\vec{s} = \frac{\sqrt{3}R}{2} i - \frac{1}{2} R\, j$

$|\vec{s}| = \sqrt{\frac{3}{4}R^2 + \frac{1}{4}R^2}$

$= \sqrt{\frac{4R^2}{4}}$

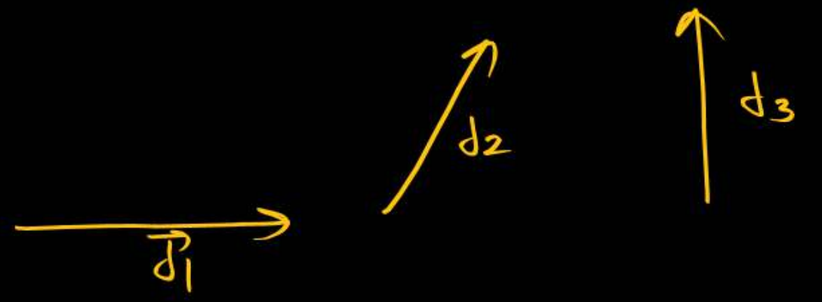
$= \sqrt{R^2} = R$

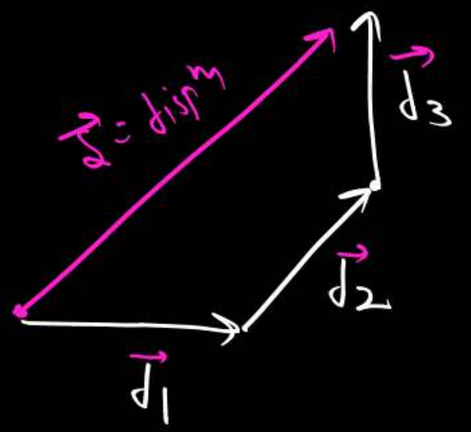
If $\theta = 60^\circ$

(26) Ramlal moves a length $\vec{d_1}, \vec{d_2}, \vec{d_3}$ as shown in fig. then find distance and displacement:-

$$\text{distance} = |\vec{d_1}| + |\vec{d_2}| + |\vec{d_3}|$$

$$\text{disp}^m = \vec{d_1} + \vec{d_2} + \vec{d_3}$$





(27)

Ramlal moves 10m North then 10m east. then Climb 10m on vertical Pole then find distance and displacement.

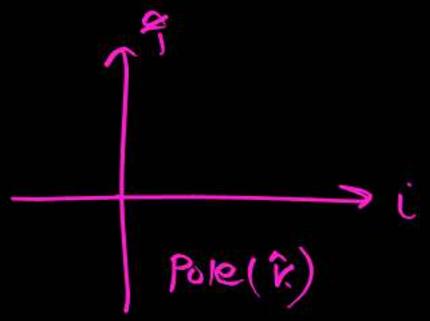
$$\text{dist}^n = 10 + 10 + 10 = 30\,m$$

$$\vec{S} = 10\hat{j} + 10\hat{i} + 10\hat{j}$$

$$|\vec{S}| = \sqrt{(10)^2 + (10)^2 + (10)^2}$$

$$= \sqrt{300}$$

$$= 10\sqrt{3}\,m$$



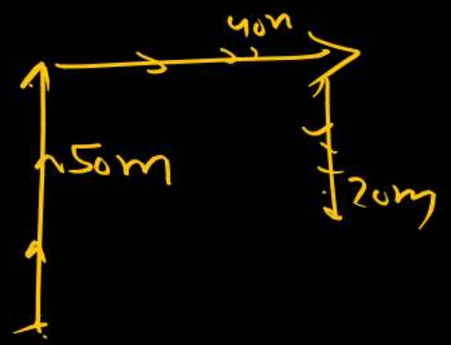
Question

(18) No Copy

A man has to go 50 m due North, 40 m due East and 20 m due South to reach the field?

(a) What distance he has to walk to reach the field? 110 m

(b) What is his displacement from his house to the field?



$\vec{s} = 50\,m$

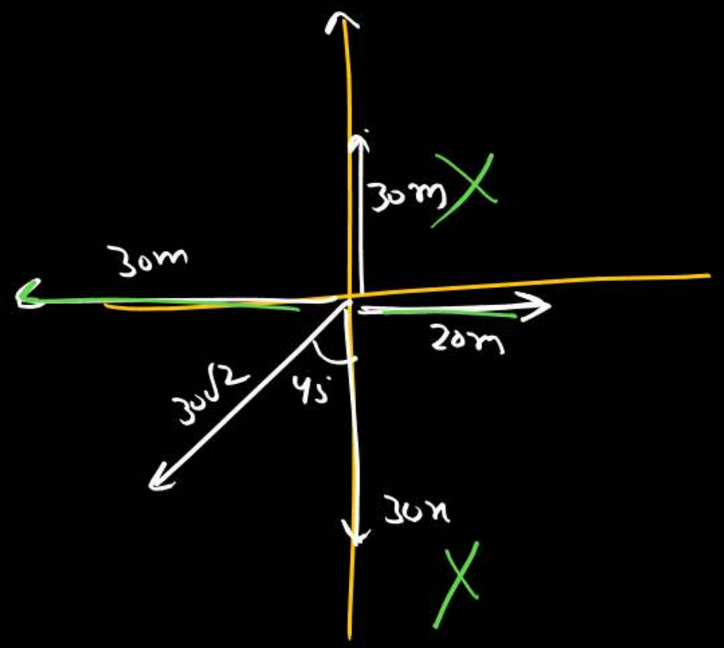
Question

(25) Cap1

A person moves 30 m north and then 20 towards east and finally $30\sqrt{2}$m in south-west, then displacement.

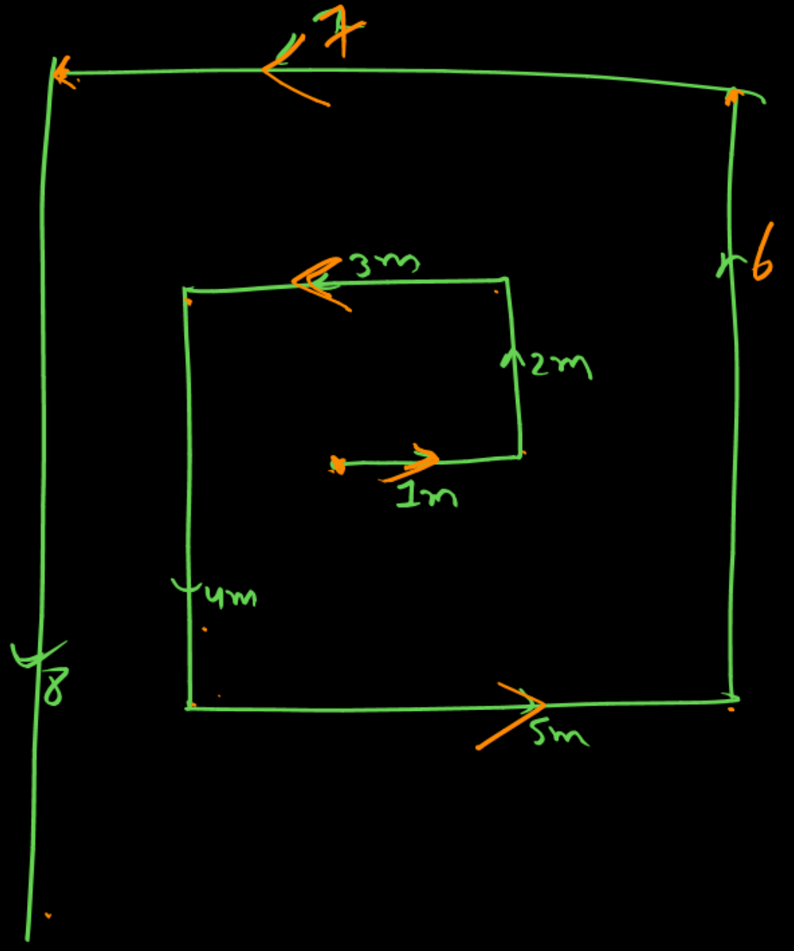
(IIT-2012)

1. 10 m North
2. 10 m South
3. 10 m West
4. Zero



$\vec{s} = 30\hat{j} + 20\hat{i} - 30\hat{i} - 30\hat{j}$

$\vec{s} = -10\hat{i}$

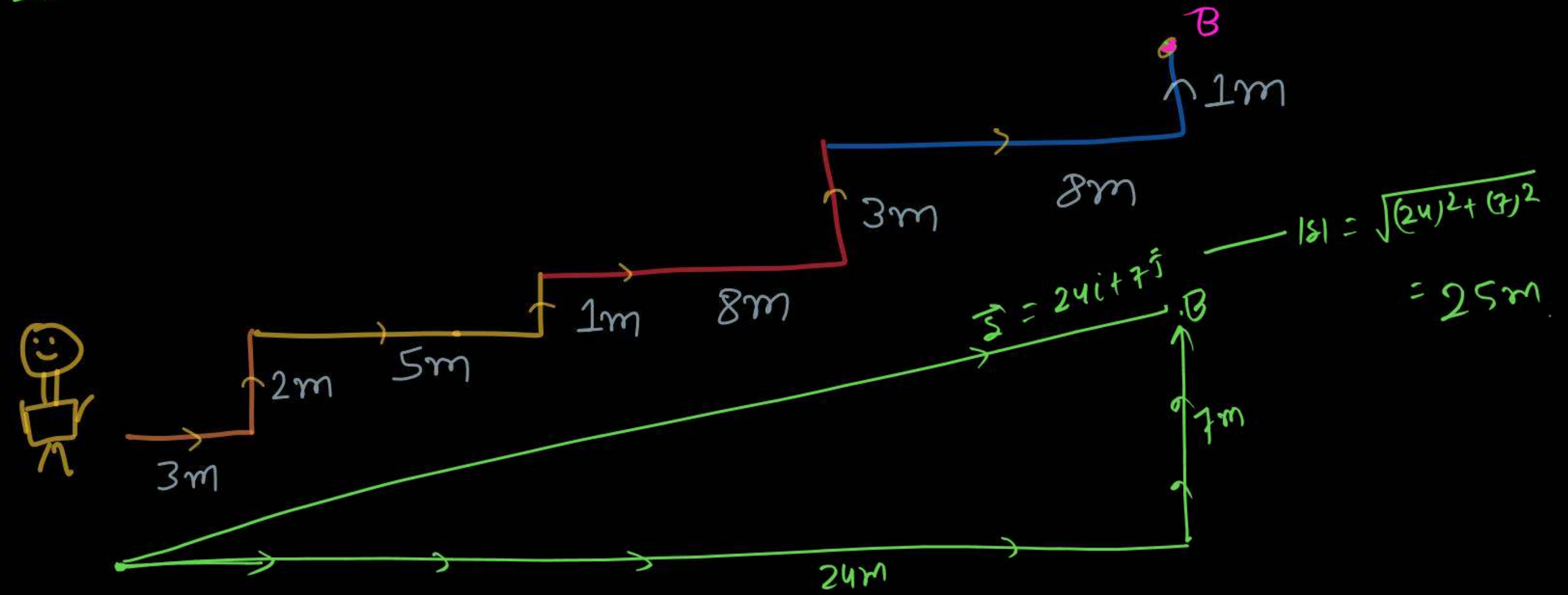


$\vec{s} = -4\hat{i} - 4\hat{j}$

$|\vec{s}| = 4\sqrt{2}$ ✓

(30)

CQ1


B
1m
8m
3m
8m
1m
5m
2m
3m
$\vec{s} = 24\hat{i} + 7\hat{j}$
B
7m
24m
$|s| = \sqrt{(24)^2 + (7)^2}$
$= 25m$

Question

31

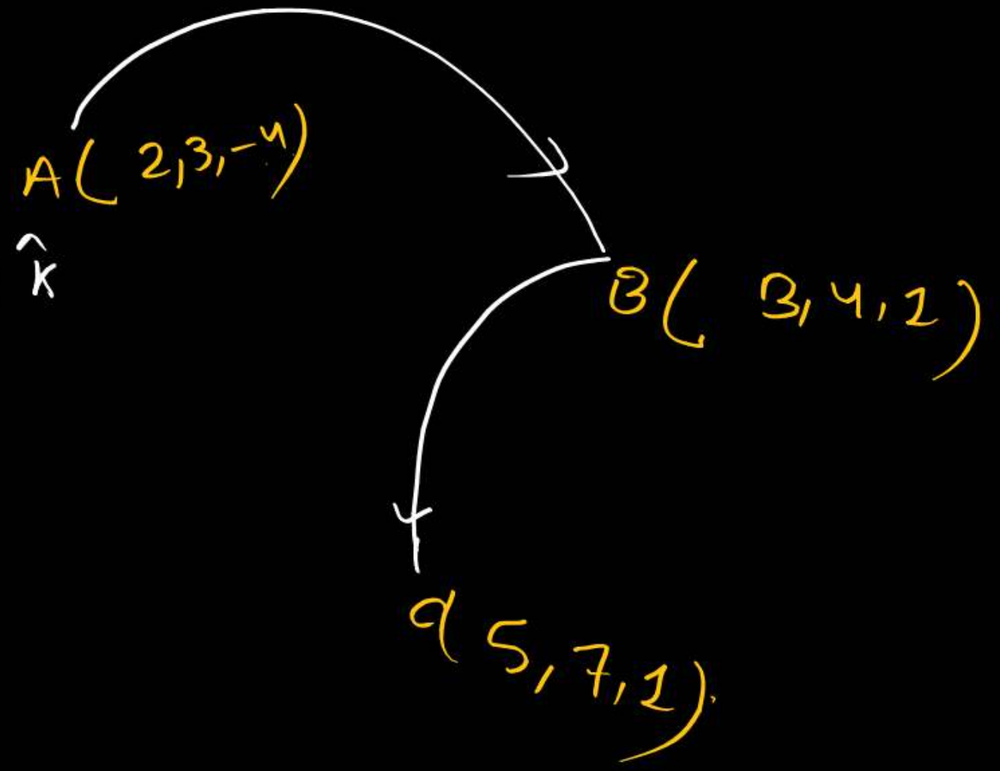
Object moves from A(2, 3, −4) to B(3, 4, 1) to C(5, 7, 1). Find distance and displacement from A to C.

dist^n = Can't find ✓

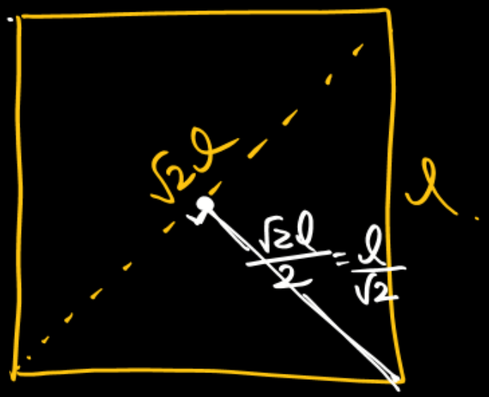
disp^n $= (5-2)\hat{i} + (7-3)\hat{j} + (1+4)\hat{k}$

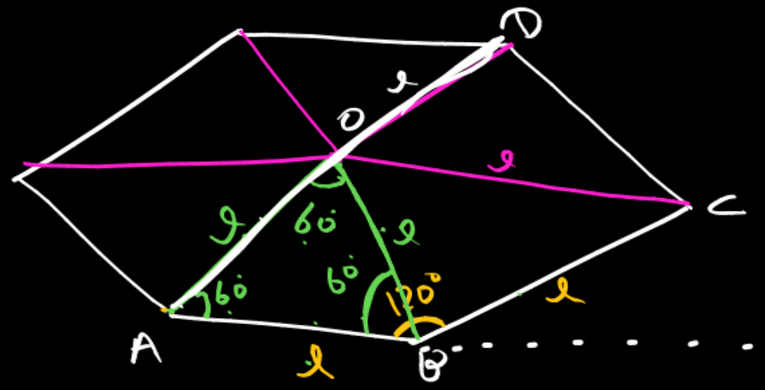
$= 3\hat{i} + 4\hat{j} + 5\hat{k}$

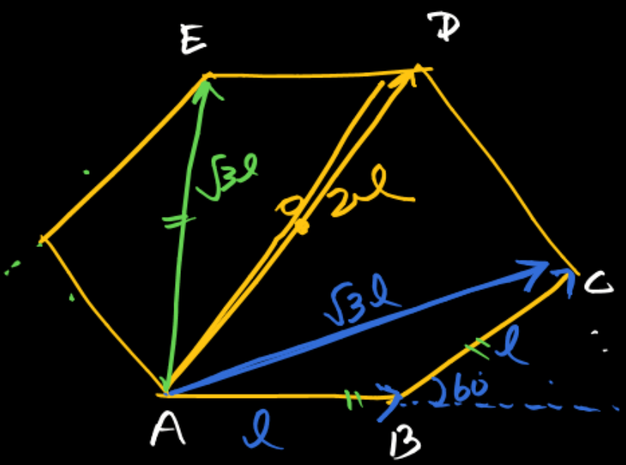
$|\vec{S}| = 5\sqrt{2}$



#

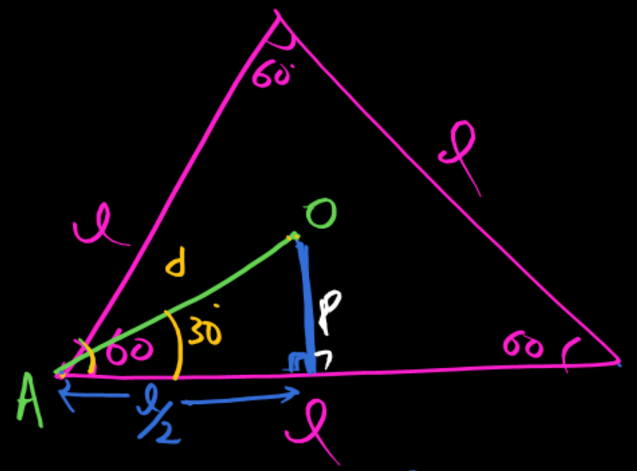






$OA = l$ = equal to side len

$AD = 2l$



$\cos 30° = \frac{\ell/2}{d}$

$\frac{\sqrt{3}}{2} = \frac{\ell/2}{d}$

$\boxed{d = \frac{\ell}{\sqrt{3}}}$

$\tan 30° = \frac{P}{\ell/2}$

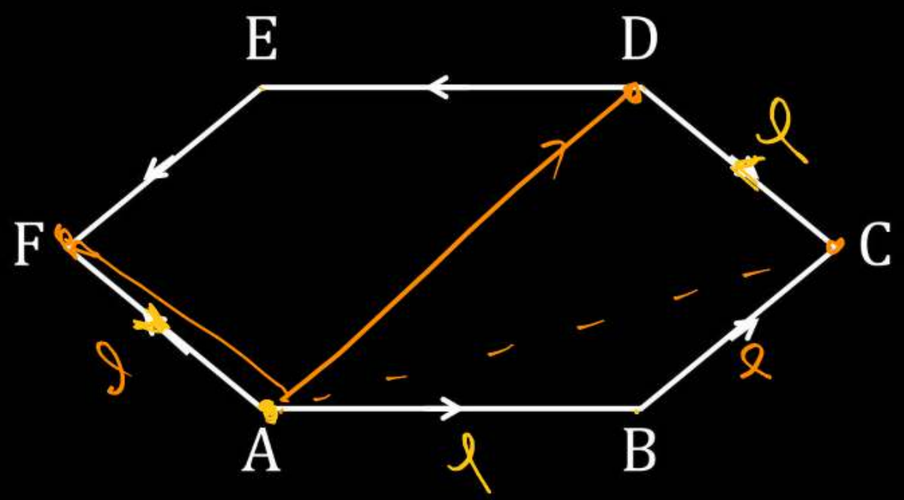
$\frac{1}{\sqrt{3}} = \frac{P}{(\ell/2)}$

$\boxed{P = \frac{\ell}{2\sqrt{3}}}$

Question

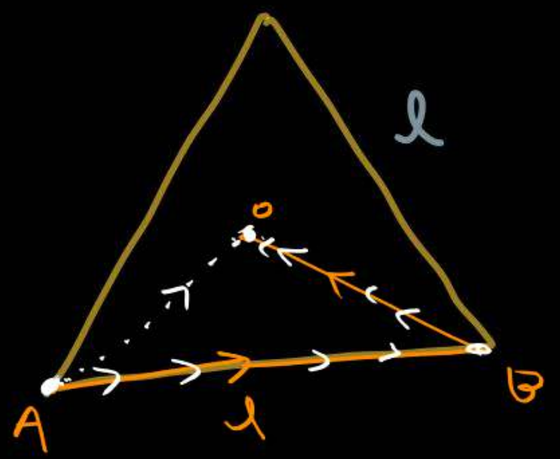
32

Object is moving on Hexagonal Path of side l.



Motion	Distance	Displacement
A → B	l	l
A → C	$2l$	$\sqrt{3}l$
A → D	$3l$	$2l$
A → E	$4l$	$\sqrt{3}l$
A → F	$5l$	l
A → A	$6l$	0

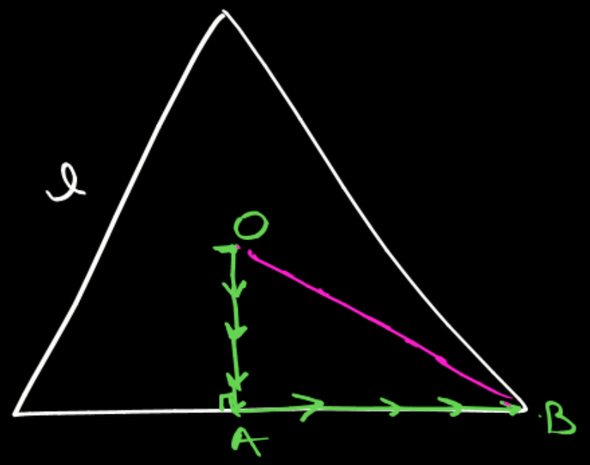
(33) Object is moving on equilateral triangle $A \rightarrow B \rightarrow O$ then find distance and displacement :-



$$\text{distance} = l + \frac{l}{\sqrt{3}}$$ ✓

$$\text{disp}^n = \frac{l}{\sqrt{3}}$$ ✓

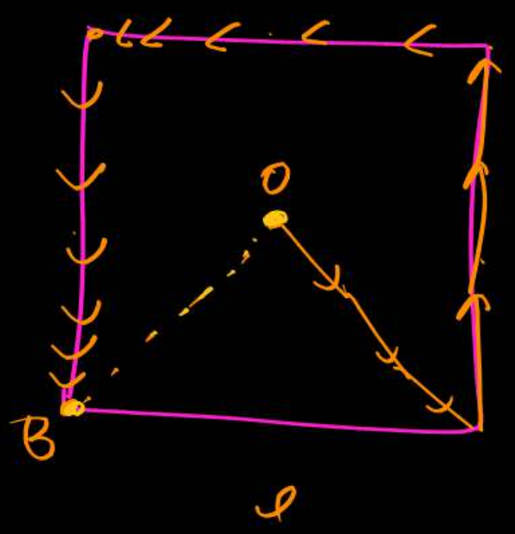
(34)



$$\text{disp}^n = \frac{l}{2\sqrt{3}} + \frac{l}{2}$$

$$\text{disp}^n = \frac{l}{\sqrt{3}}$$

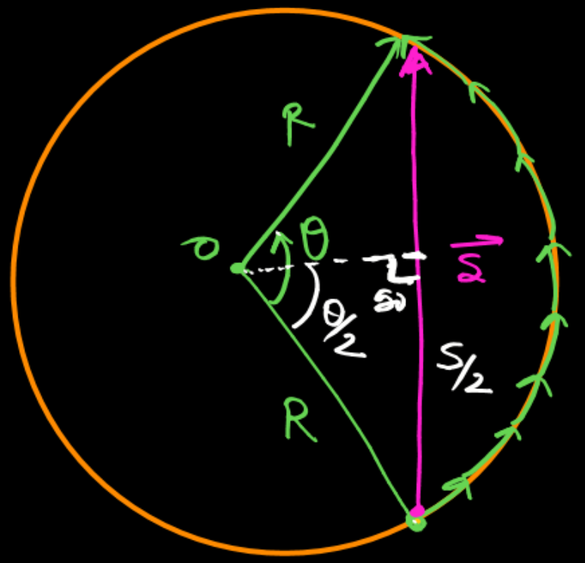
35



$$\text{dist}^n = \frac{\ell}{\sqrt{2}} + 3\ell \quad \checkmark$$

$$\text{disp}^n = \frac{\ell}{\sqrt{2}} \quad \checkmark$$

(36) Motion of Particle on circular Path :—

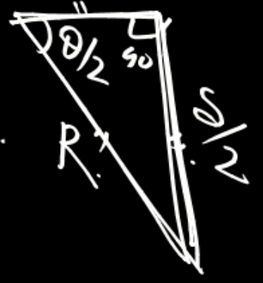


distance = ARC = $R\theta$ ✓

$\text{disp}^{m} = 2R\sin(\theta/2)$ ✓

↓ If Angle is very small $\sin(\theta/2) = \theta/2$

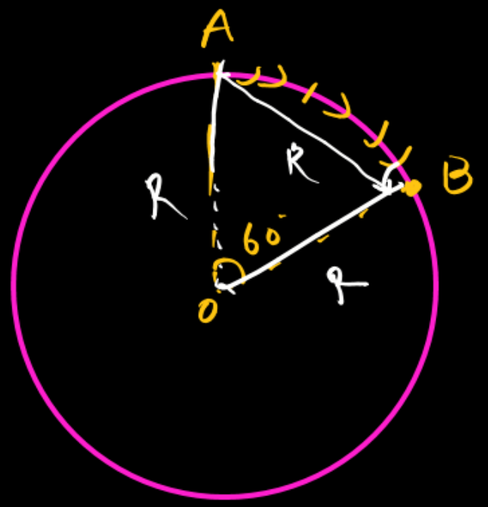
$\text{disp}^{m} = 2R \times \left[\frac{\theta}{2}\right] = R\theta$ = distⁿ



$\sin(\theta/2) = \frac{S}{2R}$

$\vec{S} = 2R\sin(\theta/2)$

(37)



$$\text{dist}^n = R\theta = R\left[\frac{\pi}{3}\right]$$

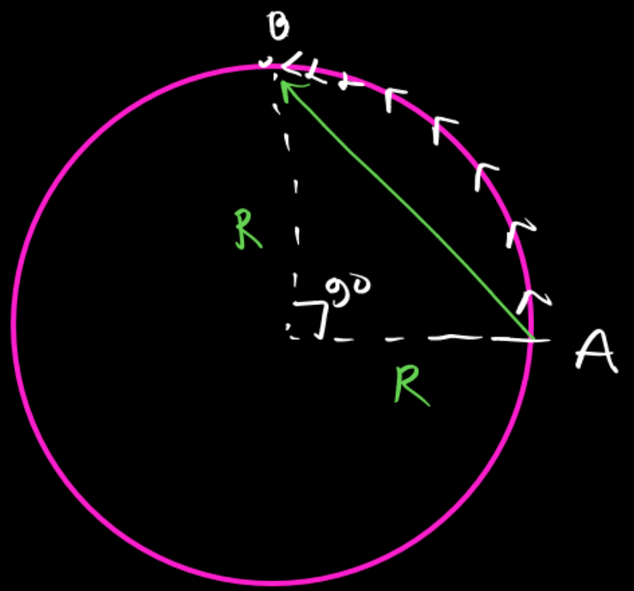
$$\text{disp}^n = 2R\sin(\theta/2)$$

$$= 2R\sin\left(\frac{60}{2}^{30}\right)$$

$$= 2R\,\frac{1}{2}$$

$$= R$$

(38)



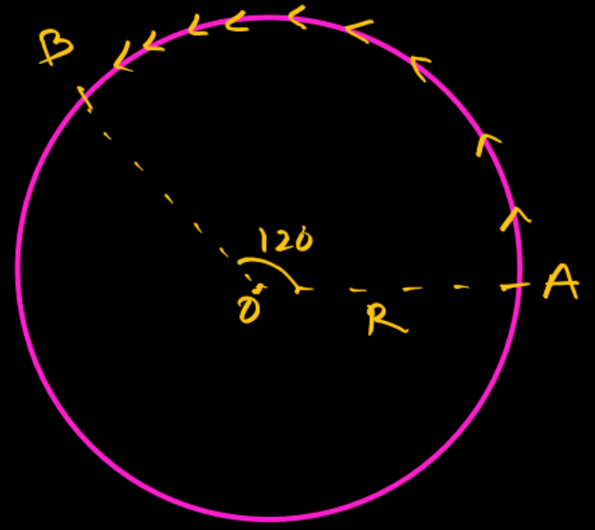
$$\text{disp}^n = R(\theta)$$

$$= R\,90^\circ$$

$$= R\frac{\pi}{2}$$

$$\text{disp}^n = \sqrt{2}R$$

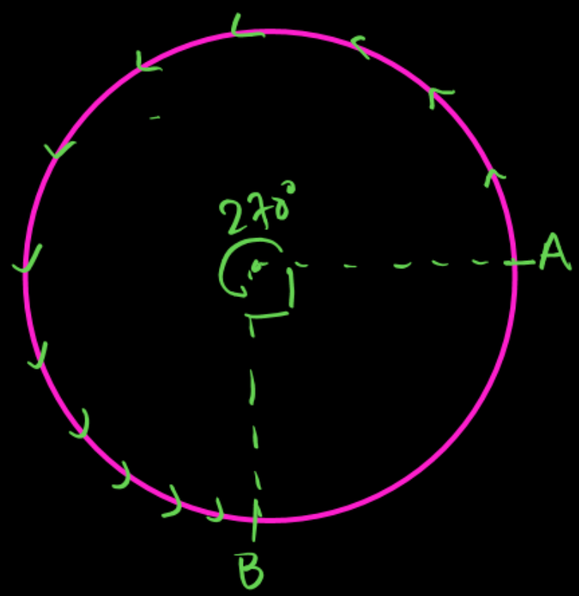
(39)



$\# \quad disp^{m} = \sqrt{3}R$

$\# \quad diston. = \frac{2\pi R}{3}$

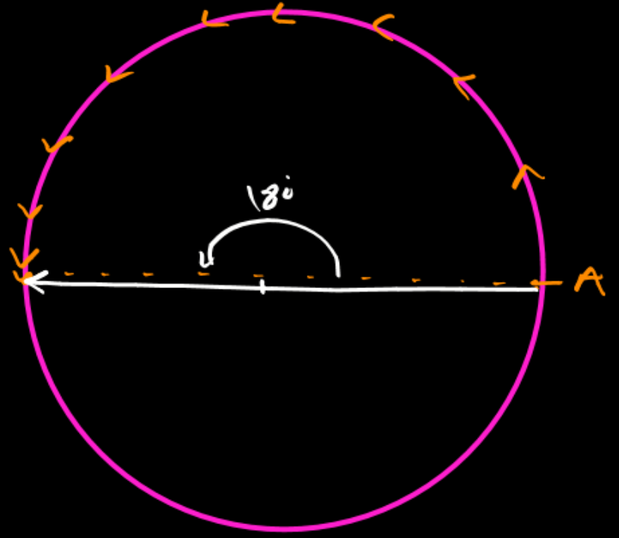
(40)



$dist^{r} = \frac{3\pi}{2}R$

$disp = \sqrt{2}R$

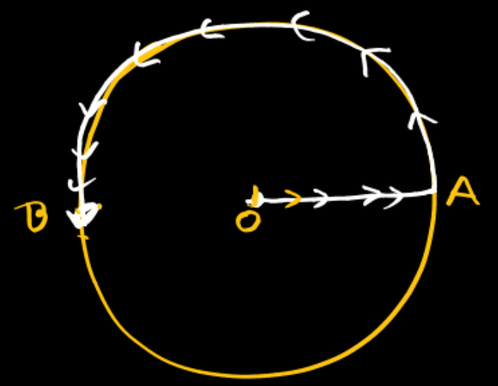
(41)



$$\text{disp}^{n} = \pi R$$

$$\text{disp}^{m} = 2R = 2R \sin\left(\frac{180}{2}\right)^{90}$$

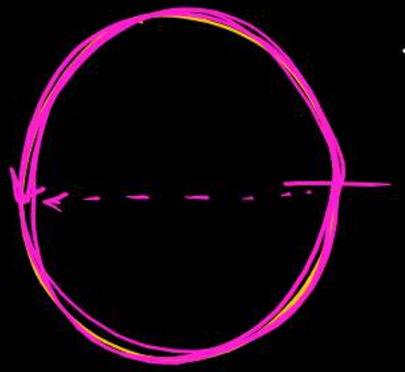
(42)



$$\text{distance}_{O \to B} = R + \pi R = R(1+\pi)$$

$$\text{disp}^{m}_{O \to B} = R$$

(43) Ram lab Compt one circul Path in 40sec then find Ratio of distance to dispⁿ in 140sec. Radius=R



T = 40sec

t = 140 = 40 + 40 + 40 + 20 (half)

$$\frac{\text{dist}^n = 7\pi R}{\text{disp}^n = 2R} = \frac{7\pi}{2}$$ ✓

(44) Ramlal moving on curve of eqn $(x-5)^2 + y^2 = 25$ then find distn when it moves from (0,0) to (10,0).

(a) ~~10m~~

(b) 5m

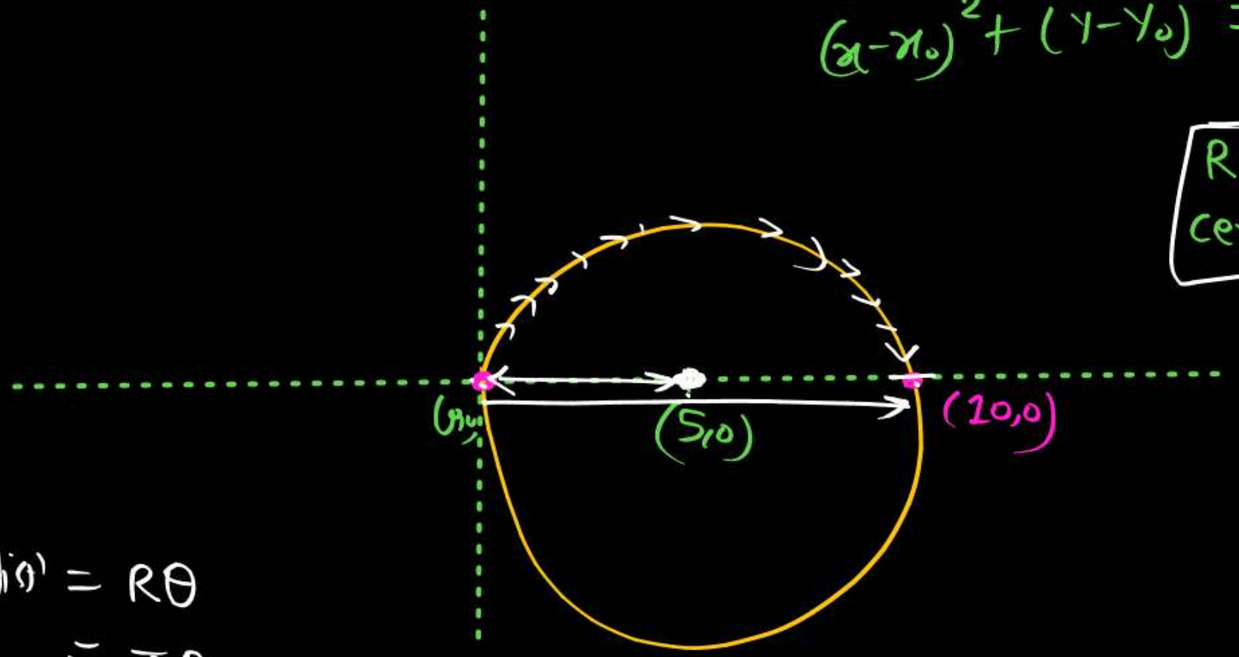
✓(c) 5π

(d) 10π

$(x-x_0)^2 + (y-y_0)^2 = R^2$

R = 5m

centre = (5,0)



dist = $R\theta$

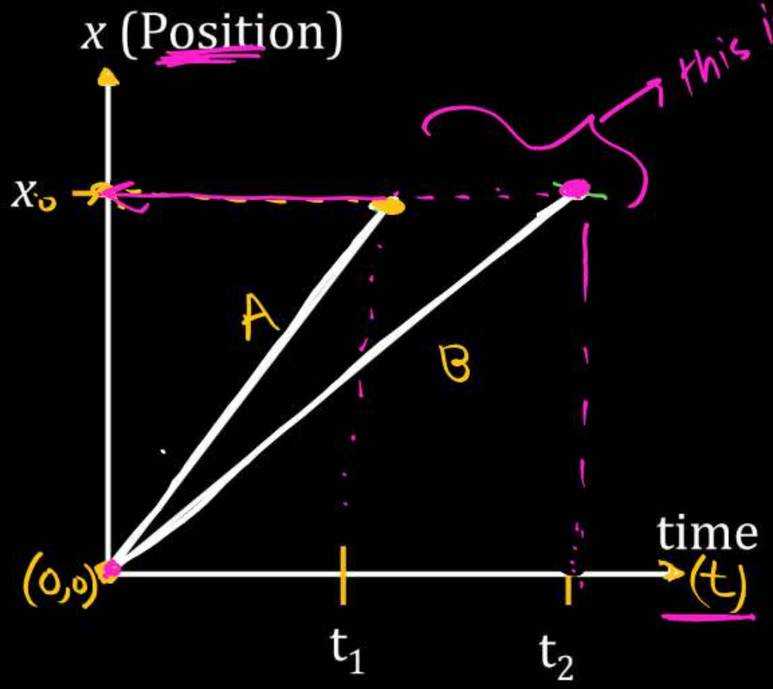
$= \pi R$

$= 5\pi$

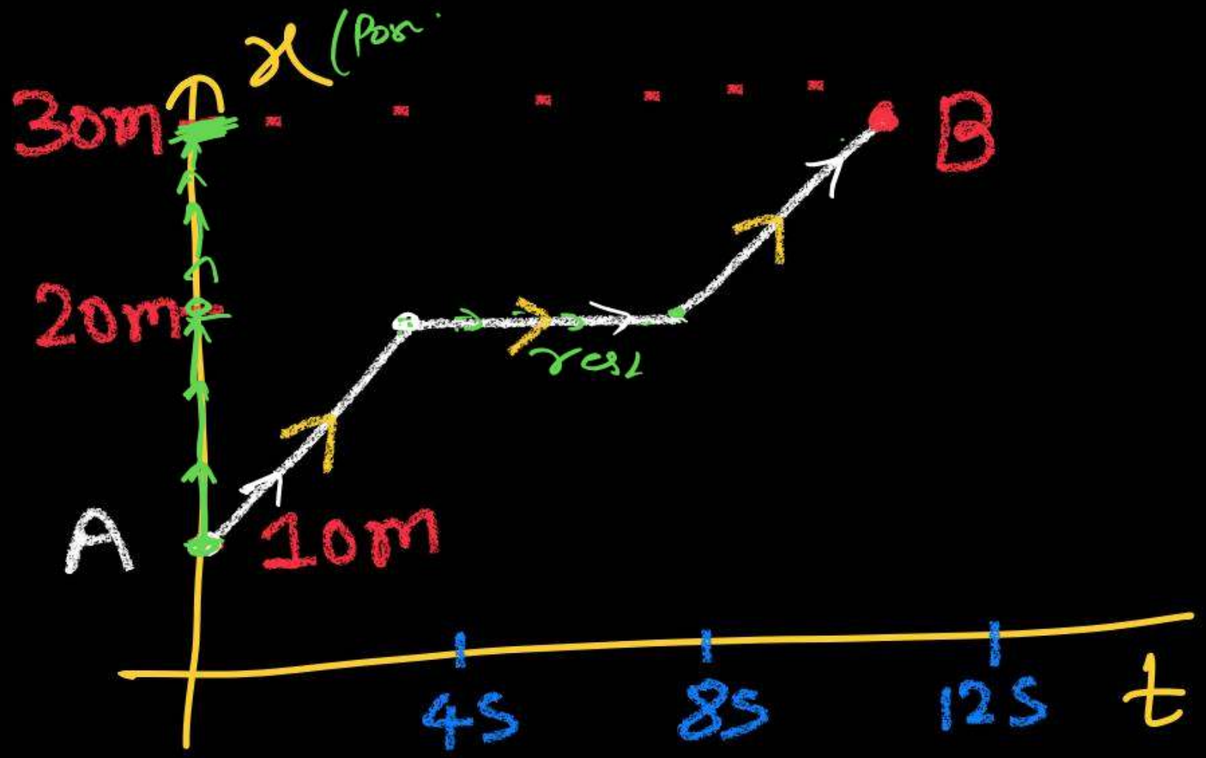
Question 42

Position-time graph for two man are moving along x-axis as shown; then find correct relation between

1. displacmeent = displament
2. displacmeent > displament
3. displacmeent < displament
4. None of these



Question 96

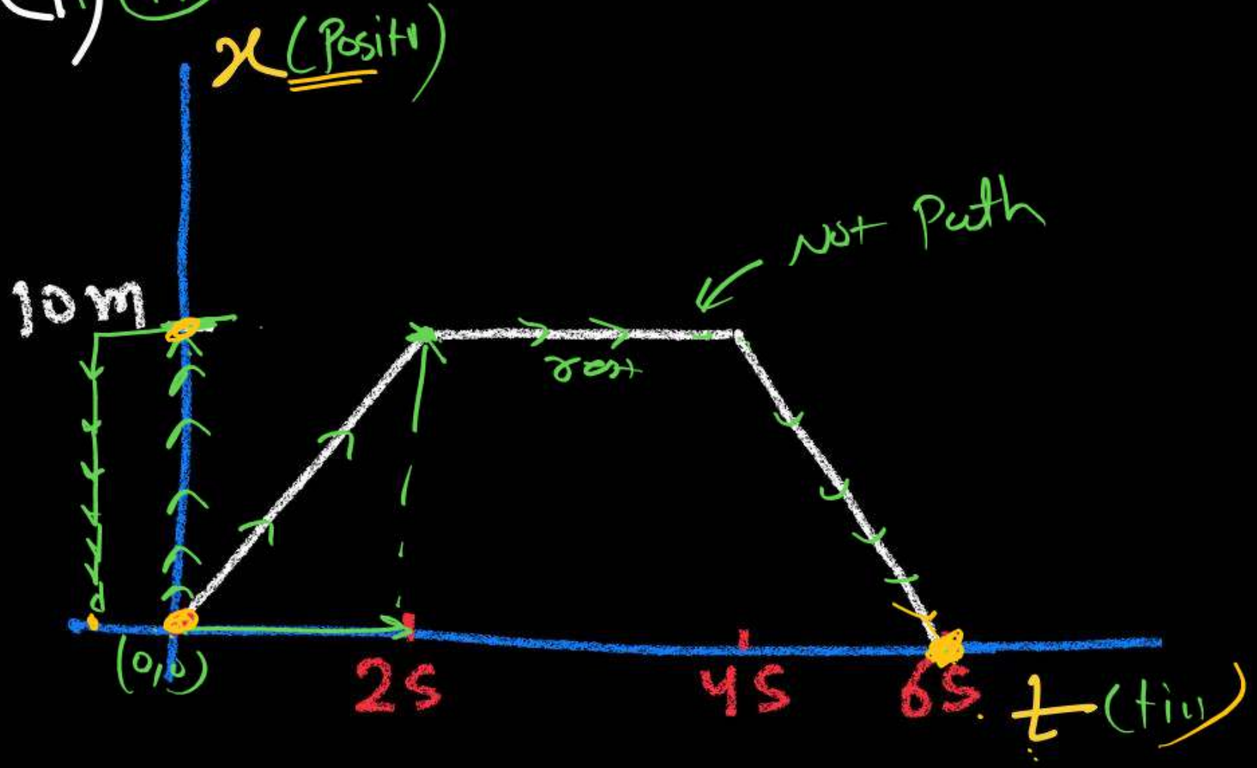


$|distance|_{A \to B} = ??$

$= 30 - 10$

$= 20\,m$

Q.14) (47)



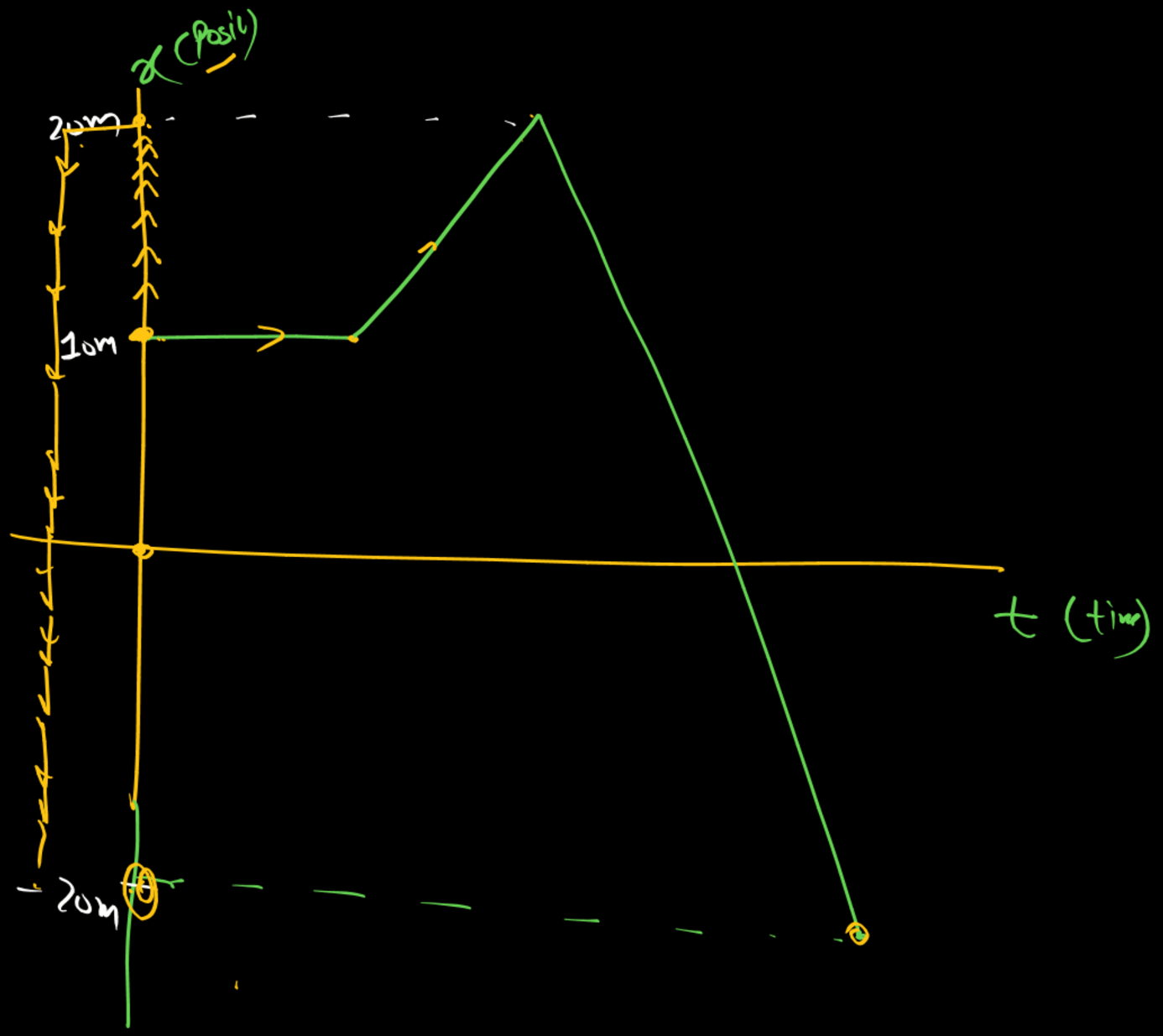
|distance|$_{A \to B}$

|disp^{m}|$_{A \to B}$

dist^{n} = 10 + 10 = 20 m

disp^{n} = 0 m

(48)



$$\text{disp}^n = -20 - 10$$
$$= -30\text{m}$$

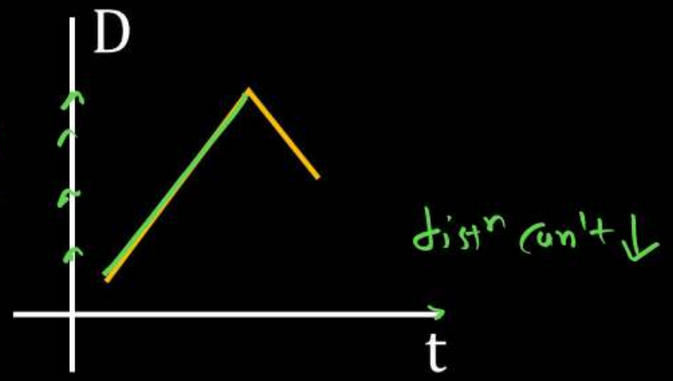
$$\text{dist}^n = 10 + 20 + 20$$
$$= \underline{50\text{m}}$$

Question

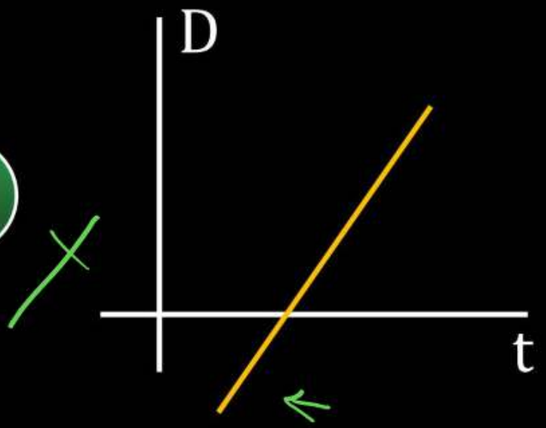
45

Which of the following graph is correct for distance-time.

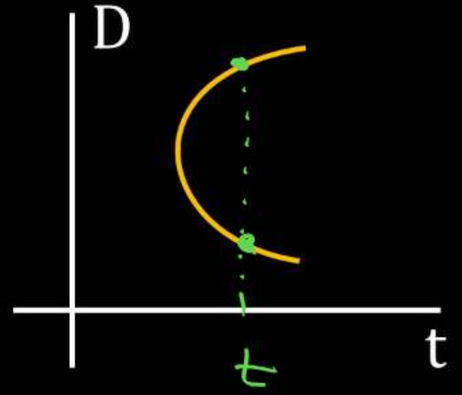
1



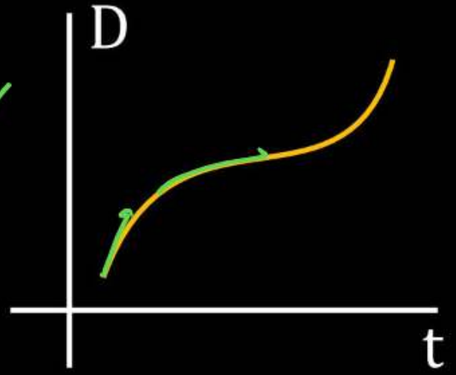
2



3

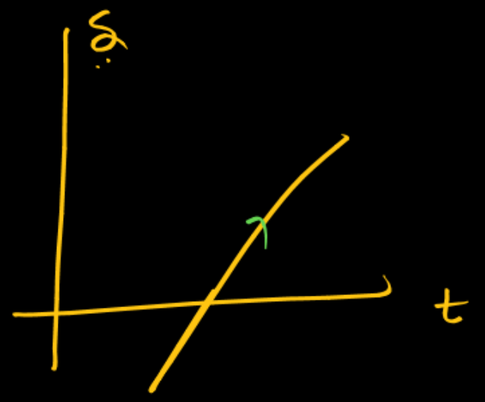


4

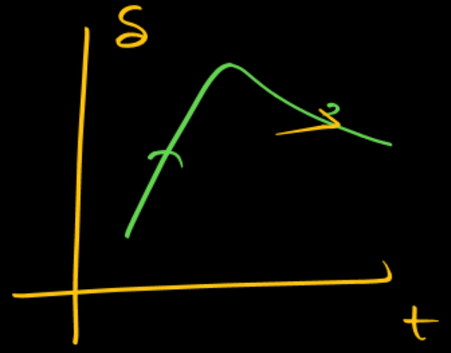


(50) Which of following Not — correct for disp^n - time graph: (22%)

(a)



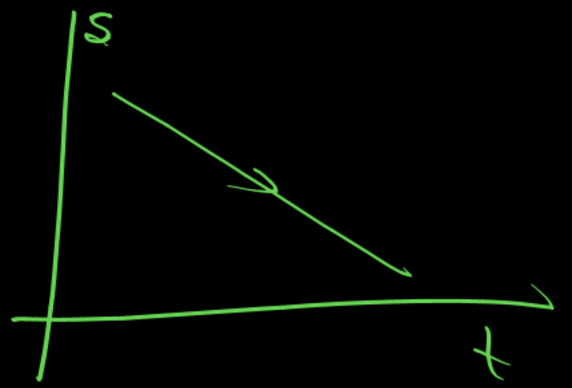
(b)



(c) Non of these

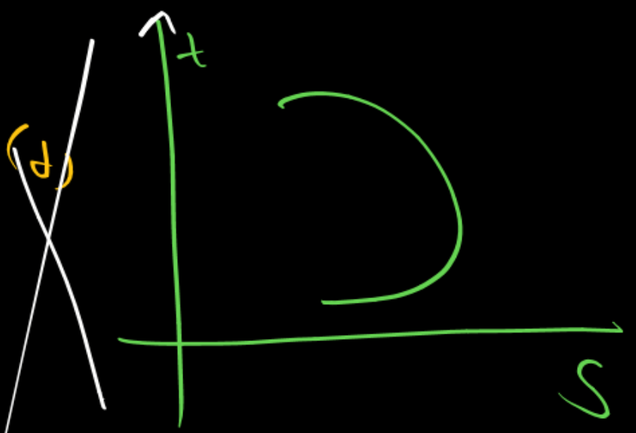
Ans. one +1

(c)



Ans 4 - are correct

(d)



Try on rough Page

Question

A butterfly start flying from a corner of the cubical room of side l and reaches to the opposite corner of the room. Find its displacement.

Question

An ant start moving from a corner of the solid cubical room and want to reach opposite of body diagonal find minimum distance moved by ant.

Question

A carrom board (4 ft × 4 ft square) has the queen at the centre. The queen, hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line. Find the magnitude of displacement of the queen

(a) From the centre to the front edge

(b) From the front edge to the hole and

(c) From the centre to the hole.

Question

A drunkard is walking along a straight road. He takes 5 steps forward and 3 steps backward and so on. Each step is 1 m long and takes 1s. There is a pit on the road 11 m away from the starting point. The drunkard will fall into the pit after:

1. 21 s
2. 29 s
3. 31 s
4. 37 s

Thank

You